Biochemistry Cume – November 14, 2014

1.(50 points) A novel peptide has been isolated from a fungus found growing in one of the restrooms at the football stadium. With funding from the athletic program, you have been asked to characterize this peptide and have decided to sequence it. Based on the results described below, what have you learned about the sequence of the peptide? Describe in detail how your answer is consistent with each of the given results.

A. Treatment with aminopeptidase and carboxypeptidase is found to result in the release of no amino acids.

B. Treatment with dabsyl chloride, fluorodinotrobenzene, or phenylisothiocyanate followed by hydrolysis yields no labeled amino acid

C. Following treatment with CNBr, the N-terminal amino acid is determined to be Ala and the C-terminal is homoserine lactone (HSL).

D. Treatment of the original peptide with trypsin and chymotrypsin under conditions which would reduce disulfide bonds yielded the following peptides

Trypsin Treatment	Chymotrypsin Treatment
Thr-Tyr-Cys-Lys	Cys-Lys-Met-Ala-Phe
Met-Ala-Phe-Cys-Lys	Cys-Arg-Leu-Phe
Leu-Phe-Cys-Arg	Cys-Arg-Thr-Tyr
Glu-Tyr-Cys-Arg	Cys-Lys-Glu-Tyr

E. Trypsin treatment of the original peptide under conditions where disulfide bonds would be intact yielded two peptides with the following composition

Thr, Tyr (2), Cys (2), Arg, Lys, Glu

And

Met, Leu, Cys (2), Phe (2) , Arg, Lys, Ala,

2.(50 points) In studies involving determination of the N-terminal amino acid or protein sequencing, it is often necessary to label the N-terminal amino group (pKa approximately 8.0) without significant labeling of Lys R-group amino groups (pKa approximately 10.8) using reagents which will react with $-NH_2$ but not with $-NH_3+$. What might be an appropriate pH for such labelling experiments? Why? **(Use calculations to justify your answer. No credit will be given without supporting calculations.)** You do not need to know anything about the reagents used for labeling other than that they react with $-NH_2$ but not with $-NH_3+$ to answer this question.